

EvoCracker

AI RESEARCH PROJECT

Phase 1 of 5 — Core System Documentation

PHASE 1

Core System Documentation

State space, agent capabilities & fitness function formulation

Deliverable Summary

- Defined the formal state space and runtime agent action space.
- Documented movement topology, node-level pathfinding state, and behavior-state transitions.
- Captured the baseline fitness formulation and adaptive playstyle multiplier logic.

Verification Checklist

- ✓ State model documented and mapped to in-project structures.
- ✓ Agent sensory/action capabilities listed in implementation terms.
- ✓ Fitness formulation documented with adaptive adjustments.

1. Game State Space

The game's state space represents the environment the AI agents navigate and the data structures that encode their current situation.

Grid-Based Environment

The world is represented as a 2D discrete grid (Grid). Each cell (GridNode) has coordinates (x, y) and contains:

- walkable — Boolean indicating if the cell can be traversed.
- weight — Movement cost derived from terrain type.
- tileType — Physical property: FLOOR_STONE, WALL, WATER, MUD, TRAP.

Navigational State

During pathfinding, each node temporarily stores search state:

- g — Exact cost from the start node.
- h — Heuristic estimated cost to the goal.
- f — Total estimated cost (g + h).
- visited, inOpenSet, depth, and parent.

Connectivity

The environment supports 4-directional (orthogonal) and 8-directional (orthogonal + diagonal) movement. Diagonal movements are cost-adjusted ($\sqrt{2}$) and prevent corner-cutting through walls.

Behavioral States

An agent's behavior is governed by a Behavior Tree transitioning between:

IDLE	PATROL	INVESTIGATE	CHASE	FLEE
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2. Agent Capabilities

Agents possess a wide array of capabilities driven by their behavior trees, genetic traits, and pathfinding systems.

2.1 Sensory Capabilities

FOV	Agents utilize a raycasting-based Field of View (FOV) to detect the player and other dynamic objects within their line of sight.
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2.2 Action Capabilities

- Movement — Pathfinding towards a specific target or exploring via patrol movements.
- Combat — Dealing damage to the player upon successful engagement.
- Cooperation — Assisting other agents to secure cooperative kills or flanking maneuvers.
- Fleeing — Retreating when at a disadvantage or executing hit-and-run tactics.

2.3 Search Algorithms

Agents use a variety of pathfinding and search algorithms. Their preference for each evolves over time:

Algorithm	Description
BFS	Complete, unweighted breadth-first search.
DFS	Depth-first traversal — exploratory, non-optimal pursuit.
IDS	Iterative deepening search combining DFS space efficiency with BFS completeness.
DLS	Depth-limited search with genome-driven depth limits.
UCS	Uniform-cost search weighted by tile movement costs.
A*	Heuristic pathfinding using $g + h$ — optimal and efficient.
Greedy BFS	Heuristic-only best-first search — fast but not always optimal.
Hill Climbing	Local heuristic improvement — fast but incomplete movement.

2.4 Genetic Traits (Genome)

Each agent possesses a Genome of values between 0.0 and 1.0 governing its capabilities:

Gene / Trait	Description
speed	Determines the agent's movement velocity.
vision	Modifies the range and arc of the agent's FOV.
aggression	Likelihood to attack versus flee.
persistence	How long the agent chases after losing line of sight.
cautiousness	Hesitation before engaging or entering unknown areas.
packTendency	Desire to group up with other agents.
ambushTendency	Preference for hiding near choke points.
patrolVariance	Degree of randomness in patrol routes.
algorithmWeights	Weights determining which of the 8 search algorithms to prefer.

3. Mathematical Formulation of the Fitness Function

The Genetic Algorithm evolves the enemy population between floors. The fitness function rewards agents that successfully challenge the player's specific playstyle.

3.1 Base Fitness Calculation

The base fitness is a weighted sum of performance metrics gathered during the agent's lifetime:

Metric	Weight	Description
Time Visible	× 0.30	Seconds the player spent in agent's FOV
Damage Dealt	× 2.00	Total damage dealt to the player
Detections	× 1.50	Times a hidden player was successfully spotted
Survival Time	× 0.20	Seconds the agent stayed alive
Area Covered	× 0.40	Unique tiles explored by the agent
Time Stuck	× -1.00	Seconds unable to move — penalized
Cooperative Kills	× 2.50	Kills/heavy damage assisted by other agents

```
Base Fitness = (TimeVisible × 0.3) + (Damage × 2.0) + (Detections × 1.5) +  
(Survival × 0.2) + (Area × 0.4) - (Stuck × 1.0) + (CoopKills × 2.5)
```

3.2 Playstyle-Adaptive Bias

A player profile is constructed (rusher, stayer, explorer, fighter, or hybrid). Base fitness is then scaled based on the agent's traits that best counter the player:

Playstyle	Adaptive Multiplier	Counter Strategy
Stayer	Persistence × 0.5	Counters players who hide frequently
Rusher	Speed × 0.5	Counters players who sprint to the exit
Explorer	Pack Tendency × 0.4	Counters players who visit every room
Fighter	Cautiousness × 0.3	Counters players who engage often
Hybrid	× 1.1 (flat)	Balanced boost when no dominant pattern

```
Final Fitness = Math.max(0, Final Fitness) // Clamped to minimum of 0
```